

Physics 201 - Lecture 27

Translational

$\vec{x}$
 $\vec{v}$
 $\vec{a}$
 $\Delta \vec{x}$

position
 velocity
 acc.
 disp.

$\rightarrow$
 $\rightarrow$
 $\rightarrow$
 $\rightarrow$

Rotational

θ
 ω
 α
 $\Delta \theta$

Newton's 2nd Law

$$\sum_i \vec{F}_i = \vec{F}_{\text{net}} = m \vec{a}$$

$\sum \vec{F}_x = m a_x$ $\sum \vec{F}_y = m a_y$



$$\tau = ?$$

$$I \alpha = \tau$$

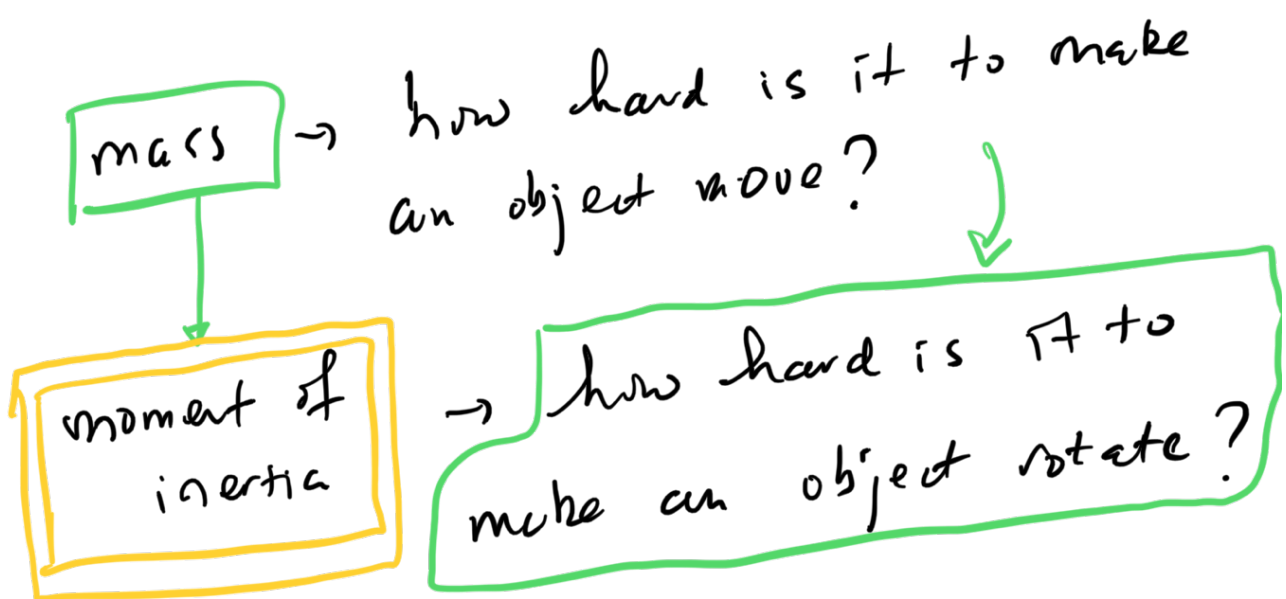
↑ moment of inertia.

... the equivalent

Question 1: What is the equivalent quantity for mass in rotational motion?

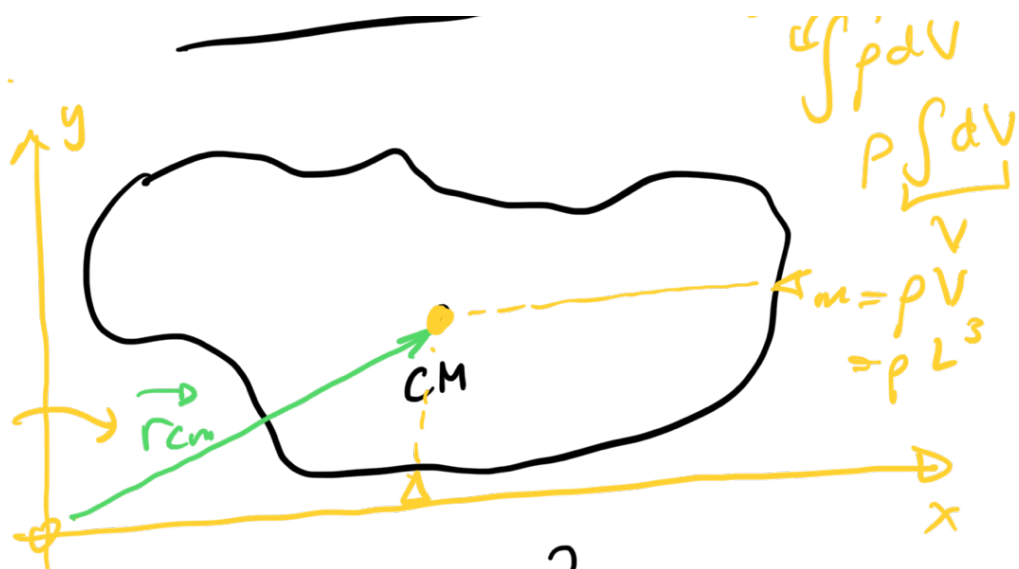
Question 2: What is the equivalent quantity for force in rotational motion?

1: Rotational Moment of Inertia



Moments





What is the total mass?

$$\underline{m} = \int_{\text{Volume}} \rho \, dV$$

density kg/m^3 (scalar)

Where is the centre of mass?

$$\underline{m} \underline{r}_{\text{CM}} = \int_{\text{Volume}} \underline{r} \rho \, dV$$

$$\underline{r}_{\text{CM}} = \frac{\int_{\text{Vol}} (\underline{r}) \rho \, dV}{\int \rho \, dV}$$

(vector)

What is the rotational moment of inertia? Well, it depends on which axis of rotation we are talking about!! Consider, for now, an axis through the center of mass.

$$m(\vec{r}_{cm})^2 = \int_{Vol} (\vec{r})^2 \rho dV$$

I

$\equiv$ rotational moment of inertia

$$m = \int_V \rho dV$$

zeroth moment of inertia.

$$m \vec{r}_{cm} = \int_V \vec{r} \rho dV$$

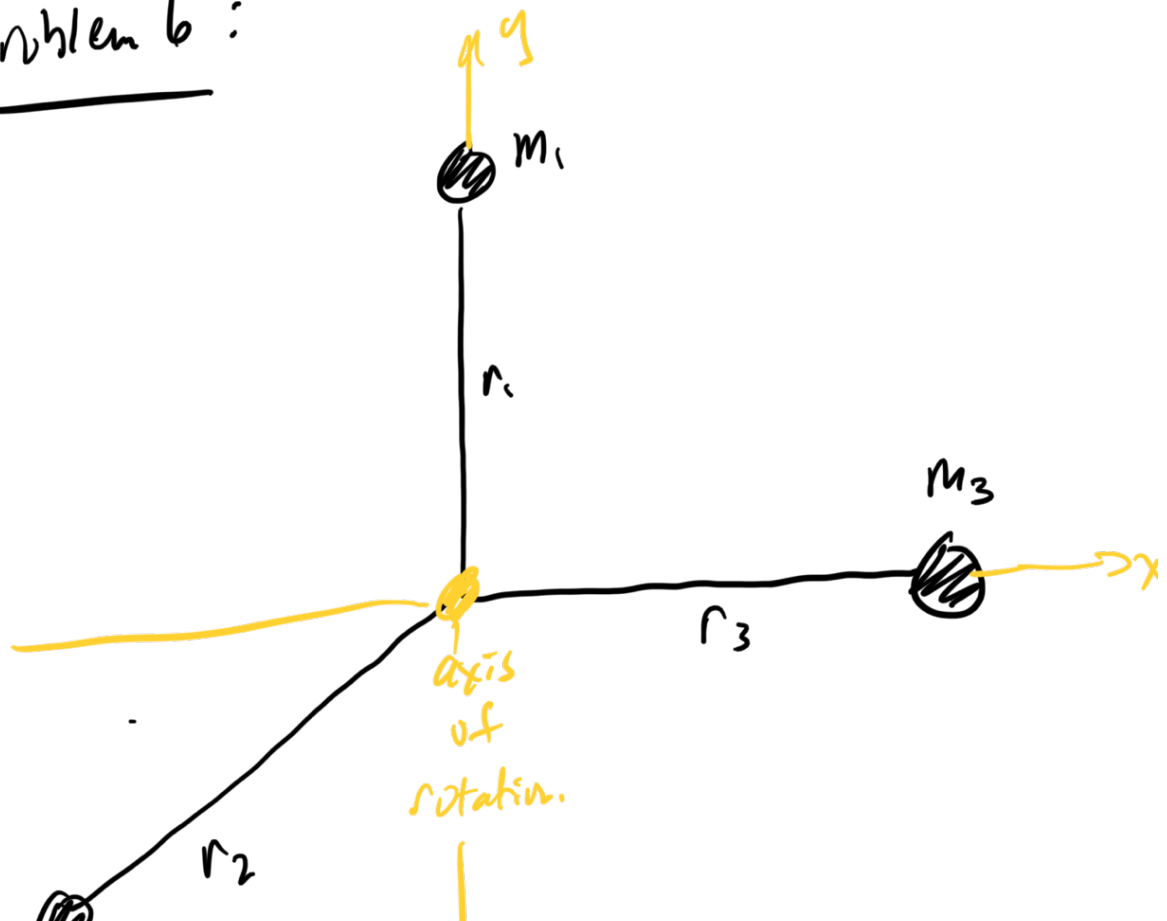
first moment of inertia

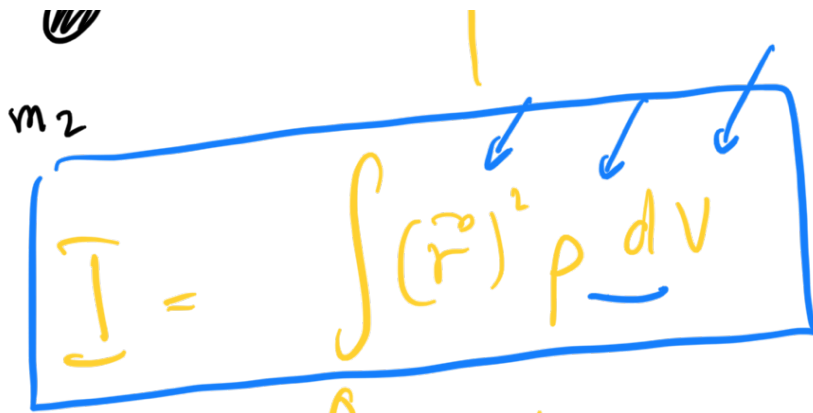
$$m(\vec{r}_{cm})^2 = \int_V (\vec{r})^2 \rho dV = \boxed{\text{Second moment of inertia.}}$$

$$m(\vec{r}_{cm})^3 = \int_V (\vec{r})^3 \rho dV = \text{third moment of inertia}$$

⋮
etc.

Problem 6:





$$I = \int (\vec{r})^2 \rho \, dV$$

$$= \int r^2 \, dm$$

$$\int \vec{r} \, dm$$

$m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3$

$\rightarrow$ "s" $\rightarrow$ sum $\rightarrow$ add up

$$= \underline{m_1 r_1^2} + \underline{m_2 r_2^2} + \underline{m_3 r_3^2}$$

$$= (.63)(.35)^2 + (.63)(.62)^2 + (.63)(.22)^2$$

$$= 0.35 \text{ kg} \cdot \text{m}^2$$

Kinetic Energy

$$K = \frac{1}{2} m v^2$$

$$\Rightarrow \boxed{R = \frac{1}{2} \overset{\downarrow}{\underset{\checkmark}{I}} \omega^2}$$

$$b) \quad \omega = 4.1 \frac{\cancel{\text{rev}}}{\text{s}} \times \frac{2\pi \text{ rad}}{\cancel{\text{rev}}}$$

$$= \underline{8.2\pi} \text{ rad/s}$$

$$R = \frac{1}{2} (I) (\omega^2)$$
$$= \frac{1}{2} (.35) (8.2\pi)^2$$

$$= 116 \text{ J}$$

Question: What is the moment of inertia of a wheel, a cylinder,

a sphere, a meter stick, etc.

There are all common shapes, and
someone has already done the
calculus for us! 😊

$$I_{\text{disk/wheel}} = \frac{1}{2} MR^2$$



$$I_{\text{sphere}} = \frac{2}{5} MR^2$$



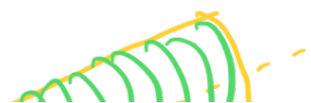
$$I_{\text{rod}} = \frac{1}{12} ML^2$$

(about center)



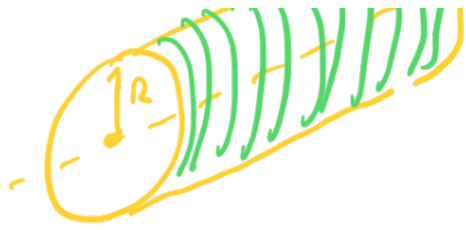
$$I_{\text{rod}} = \frac{1}{3} ML^2$$

(about one end)

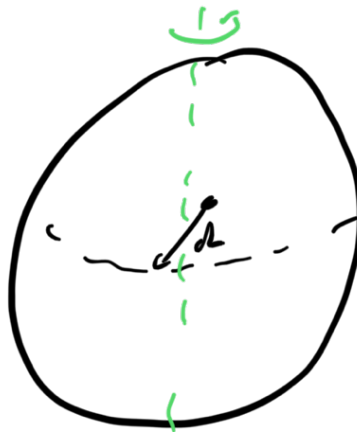


end).

$$I_{\text{cylinder}} = \frac{1}{2} MR^2$$



Question 7:



1 rev in
one "day"
 $\uparrow$
 T

$$I_{\text{sphere}} = \frac{2}{5} MR^2$$

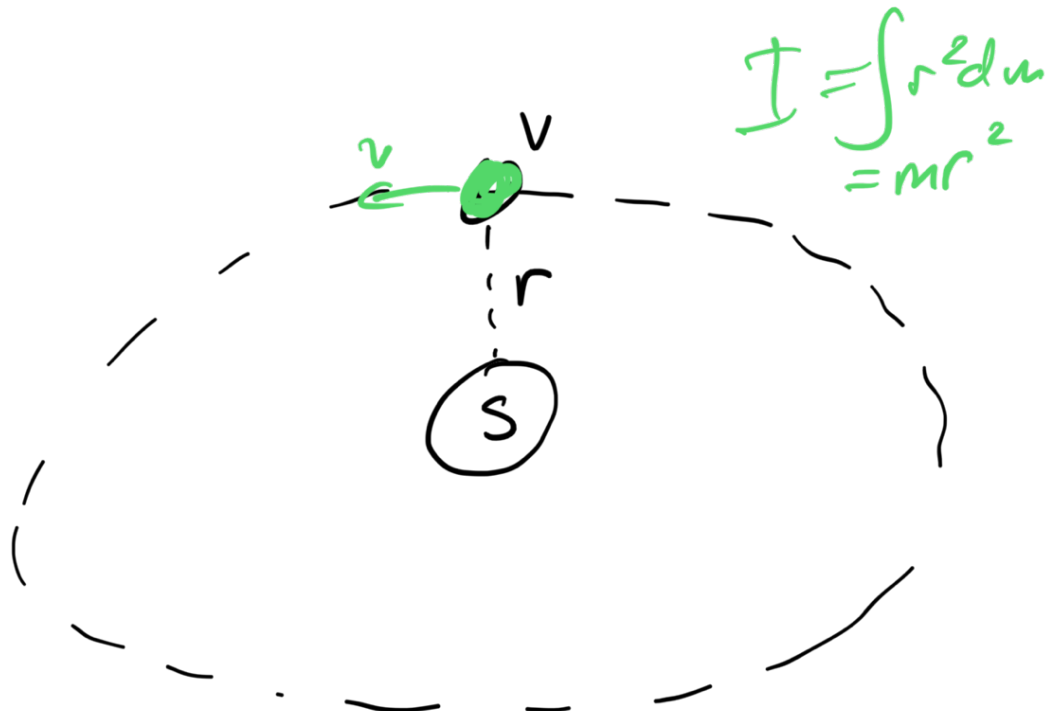
$$\omega = \frac{2\pi}{T} \text{ rad}$$

$$\begin{aligned} a) R_{\text{KE}} &= \frac{1}{2} I \omega^2 \\ &= \frac{1}{2} \left(\frac{2}{5} MR^2 \right) \left(\frac{4\pi^2}{T^2} \right) \end{aligned}$$

$$\begin{aligned}
 &= \frac{4\pi^2}{5} M \frac{R^2}{T^2} \\
 &= \frac{4\pi^2}{5} \left(4.9 \times 10^{24} \right) \left(\frac{6.05 \times 10^6 \text{ m}}{5830 \times 3600} \right)^2 \\
 &= \boxed{3.21 \times 10^{24}} \text{ J}
 \end{aligned}$$

$D = 12100 \text{ km}$
 $R = 6050 \text{ km}$
 $= 6.05 \times 10^6 \text{ m}$

b)



$$I = \underline{\underline{mr^2}} \qquad \omega = \underline{\underline{\frac{2\pi}{T}}}$$

$$K = \frac{1}{2} I \omega^2$$

$$= \frac{1}{2} (m r^2) \left(\frac{2\pi}{T} \right)^2$$

$$= \frac{2\pi^2 m r^2}{T^2}$$

$$= 2\pi^2 \left(4.9 \times 10^{24} \right) \left(\frac{1.08 \times 10^4}{225 \times 24 \times 3600} \right)^2$$

$$= 2.98 \times 10^{33} \text{ J}$$

Back to Newton's 2nd Law for rotational motion

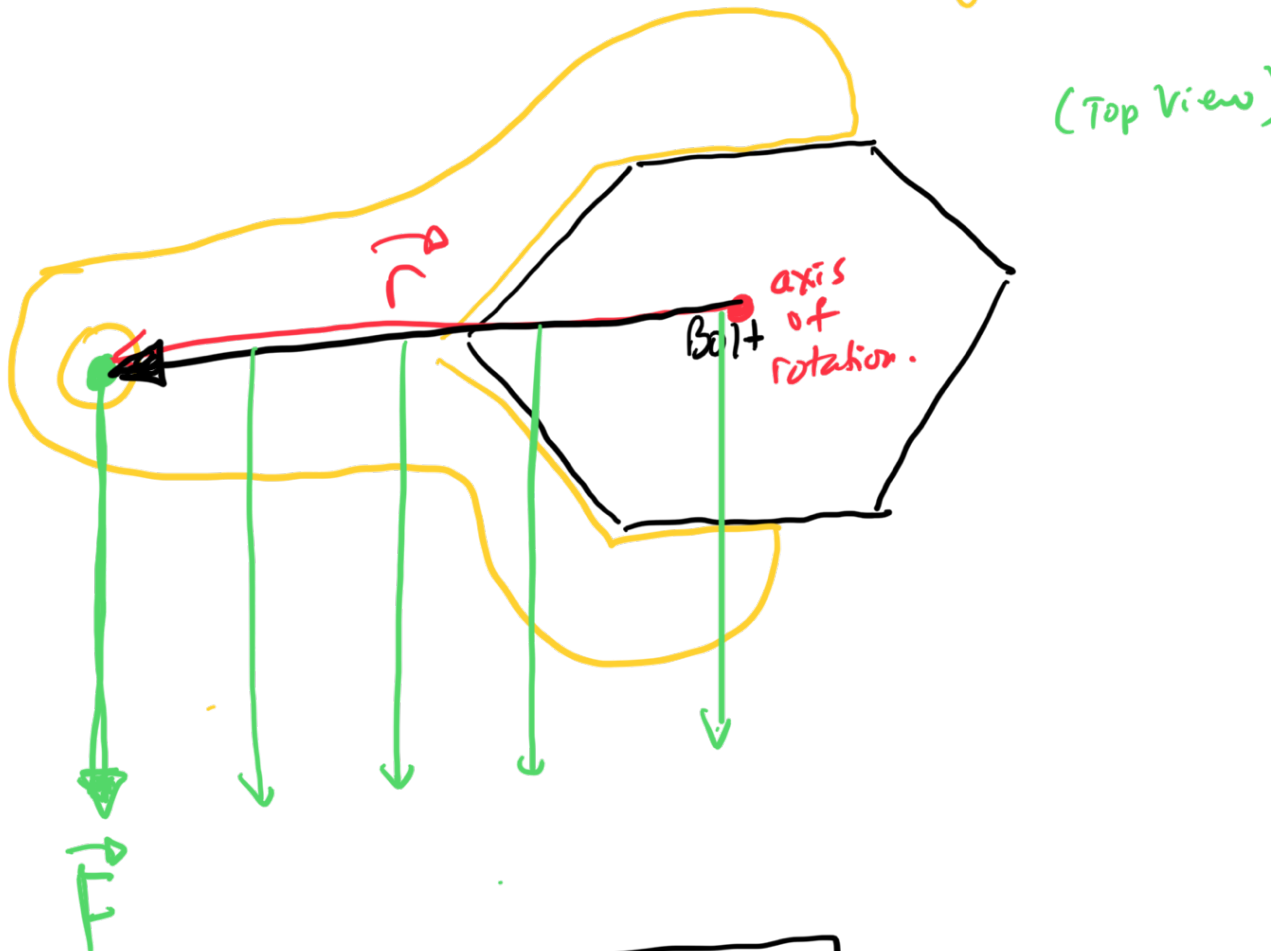
$$\tau = I \alpha$$

What is the rotational equivalent of accelerate linearly . . .

force? Force $\rightarrow$ acceleration
Torque $\rightarrow$ angular acceleration

$\Rightarrow$ **Torque** (τ)

Stupid IKEA wrench



$\vec{F}$ $\equiv$ applied force

$\vec{r}$ $\equiv$ vector which points from the axis of rotation to the point of application of the

- point of application of force.

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$\vec{r}$ cross $\vec{F}$

Torque is a vector!

$\vec{r} \times \vec{F}$ is the vector / cross product of $\vec{r}$ and $\vec{F}$. We need to understand how to calculate both the size and direction of this torque vector.

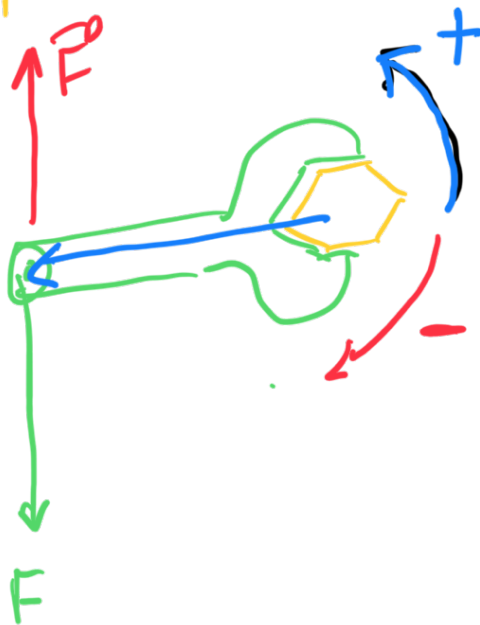
$$W = \vec{F} \cdot \Delta \vec{x} \\ = |\vec{F}| |\Delta \vec{x}| \cos \theta$$

size:

$$|\vec{\tau}| = \tau = |\vec{F}| |\vec{r}| \sin \theta$$

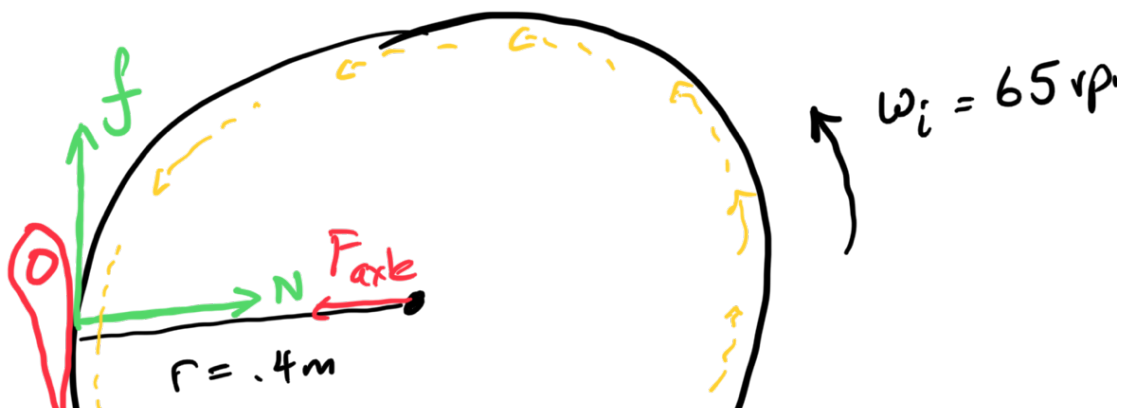
direction:

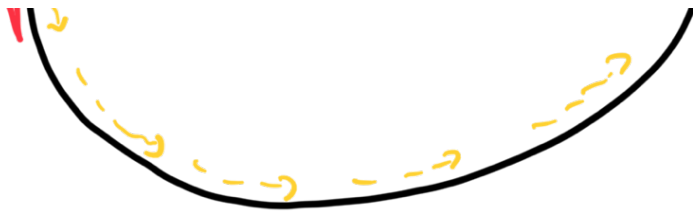
- Think about which direction of rotation the torque will tend to cause.



∴ this
is a positive
torque.

Question 8:





$$N = 14 \text{ N}$$

$$\begin{aligned} f &= \mu_k N \\ &= (0.40)(14) = 5.6 \text{ N} \end{aligned}$$

$$\begin{aligned} \tau_f &= -f \cdot r \cdot \sin(90^\circ) \\ &= -(5.6)(.4)(1) \end{aligned}$$

$$\tau_f = -2.24 \text{ N}\cdot\text{m}$$

$$\tau_N = N \cdot r \cdot \sin(180^\circ) = 0 !$$

$$\tau_{F_{\text{axle}}} = (F_{\text{axle}})(0) \sin(0^\circ) = 0 !$$

$$\begin{aligned} \circ \circ \tau_{\text{NET}} &= \tau_f + \tau_N + \tau_{\text{Faxle}} \\ &= -2.24 \text{ N}\cdot\text{m} \end{aligned}$$

$$\begin{aligned} I &= \frac{1}{2} m r^2 && (\text{disk}) \\ &= \frac{1}{2} (9.1) (.4)^2 \\ &= 7.28 \text{ kg}\cdot\text{m}^2 \end{aligned}$$

$$\begin{aligned} \tau_{\text{NET}} &= I \alpha \\ -2.24 &= 7.28 \alpha \end{aligned}$$

$$a) \quad \alpha = -0.3077 \text{ rad/s}^2$$

$$b) \quad \omega_i = 65 \text{ rpm} = 65 \frac{\text{rev}}{\text{min}} \times \frac{2\pi \text{ rad}}{\text{rev}}$$

$$\times \frac{1 \text{ min}}{60 \text{ s}}$$

$$= 6.8068 \text{ rad/s}$$

$$\omega_f = 0$$

$$\alpha = -0.3077 \text{ rad/s}^2$$

$$\Delta\theta = ?$$

$$\cancel{\omega_f^2} = \omega_i^2 + 2\alpha\Delta\theta$$

$$0 = (6.8068)^2 - 2(.3077)\Delta\theta$$

$$\Delta\theta = 75.288 \text{ rad}$$

$$\# \text{ rev} = \frac{\Delta\theta}{2\pi} = 11.98 \text{ rev}$$

$$\sim 12 \text{ rev.}$$

